

► Image registration

From theory to practice

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Image registration

From theory to practice

1 Introduction

Image registration is the process of aligning two or more images of the same scene. This process involves designating one image as the reference (also called the fixed image), and applying geometric transformations to the other images (moving image) so that they align with the reference.

Geometrical transformation is determined to align points in one view of an object with corresponding points in another view of the same or another object.

In general, a basic registration process includes the following phases:

1. Feature detection: detect and describe keypoints
2. Feature matching: establish keypoint correspondence between images
This can be performed manually or automatically
3. Transform model estimation: determine the geometric transform, using the correspondence
4. Image resampling and transformation

You can use image registration to align medical images captured with different diagnostic modalities (MRI, CT, PET, US and PET). Image registration allows you to compare common features in different images. It enables combining morphological and functional information. For example, you might discover a tumour which is visible in MRI and combine it with a PET image. In this type of medical applications, the images are often 3D, and so is the registration. However, image registration is also useful in 2D images, for instance, X-ray fluoroscopy, or in endoscopic videos. Here, the purpose of image registration could be the compensation of motion in sequential video frames. Images can be misaligned for a variety of reasons.

Registration covers the following application groups, according to difference in acquisition:

1. View: images of the same scene are taken from different viewpoints. Applications include mosaicking of images and shape recovery.
2. Time: Images are taken at different times under other circumstances, such as for monitoring therapy effect.

3. Sensors: images are acquired by different sensors and therefore information can be combined, such as the previously mentioned PET/MRI combination.
4. Model to scene: A model is matched with the image for integration and combination, such as specimen classification combined with image information.

This can be implemented in biopsy planning, cancer staging, evaluating treatment response.

The aim of the syllabus is to understand the theory behind image registration, common methods, and applications.

The following topics will be discussed in the following subsections: geometrical transforms, rigid and non-rigid methods, matching criteria, optimization, sampling and practical implementation using software packages for registration.

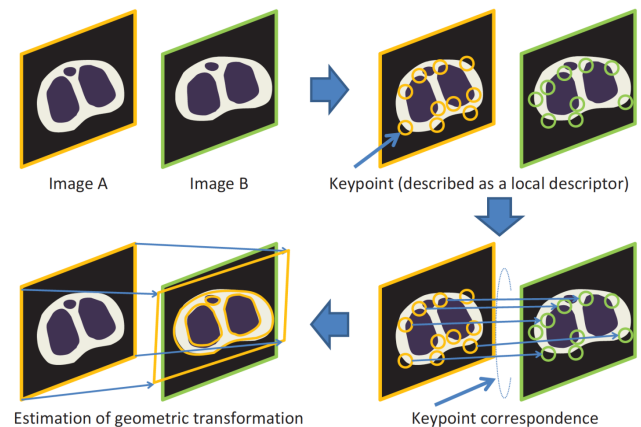


Figure 1: Registration phases

In Figure 2 an overview of a basic one stage registration framework is shown. The subtopics will be discussed below in more detail.

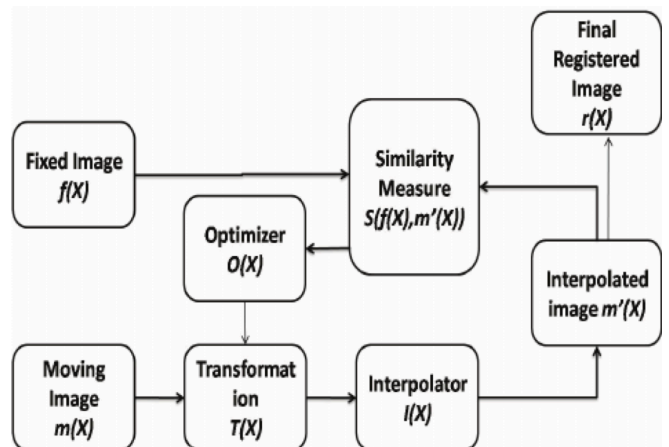


Figure 2: Basic overview of components registration process

Fixed and moving image

The basic input involves two images: a fixed image $f(X)$ and a moving image $m(X)$, where X represents a position in N -dimensional space. In an iterative process registration can be seen as an optimization problem, in which the goal is to align the moving image with the fixed image and find the spatial mapping to reach it.

Iterative process and performance measure

T (X): transform component, the spatial mapping of points from the fixed image space to points in the moving image space.

I (X): interpolator is important for evaluation of the moving image intensities at positions independent of the grid.

S (f, m•T): metric component, performance measure and indicates the quality of matching of the transformed moving image with the fixed image.

O (X): optimizes the metric component, and the search space is defined by the parameters of the transform.

$$x = a \rho \cos(b\phi)$$

$$y = a \rho \sin(b\phi)$$

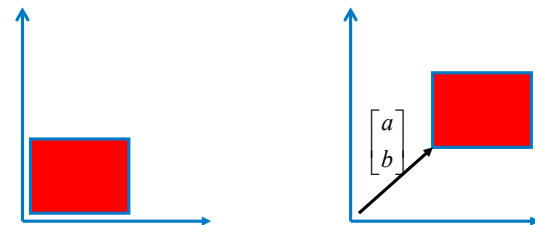


Figure 3: Translation

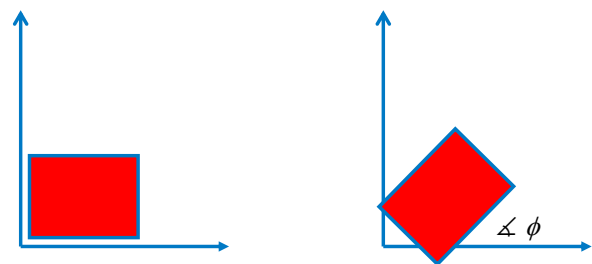


Figure 4: Rotation

2 Geometrical transforms

In the registration process each view is referred to a coordinate system, defining the space for that specific view. Transformations are divided in two categories: rigid and non-rigid transformations. Rigid transformations do not change the shape and size of the pre-image, while non-rigid transformations do change the size, while the shape is similar. The transformations can be subdivided in linear and non-linear registrations. Linear registrations contain linear transformations with translations such as rigid (translations + rotations), rigid + scaling and affine. The non-linear registration includes non-rigid, deformable, elastic and fluid.

2.1 Linear registration

The geometrical transformation is specified by two components: a translation and rotation. The **translation**, a horizontal and/or a vertical shift, is the most basic linear transformation.

$$x' = x + A$$

$$y' = y + B$$

Rotation and scaling such as zooming in are other transformations as shown in Figure 4. The rotation transformation is given by:

The **Euclidean** transform only rotates and translates. As such, shapes, sizes, angles, and distances are preserved.

$$x = x \cos(\theta) - y \sin(\theta) + a$$

$$y = x \sin(\theta) + y \cos(\theta) + b$$



Figure 2: Euclidean transform

The **similarity** transform rotates, translates, and scales isotropically. Shapes are preserved.

$$x = s x \cos(\theta) - s y \sin(\theta) + a$$

$$y = s x \sin(\theta) + s y \cos(\theta) + b$$



Figure 5: Similarity



Figure 6: affine transform

Affine transformations are a combination of rotation, scaling, shear and their combinations, the transforms are given by the equations:

$$x' = Ax + By + C$$

$$y' = Dx + Ey + F$$

Note that this transform can change the shapes, the angles between lines, and the distances between points. However, lines that are parallel in the source domain are still parallel in the destination. If we have two angles in the source domain that are equal, then they are also equal in the destination domain.

Special cases occur for different choices of the 6 parameters A, B, C, D, E and F .

Although not linear, the projective transform is often mention in the same group of the transforms since in homogeneous coordinates, it turns into a linear transform. The equations are given by:

$$x' = \frac{Ax + By + C}{Gx + Hy + 1}$$

$$y' = \frac{Dx + Ey + F}{Gx + Hy + 1}$$



Figure 3: Projective transform

Note that, from all the linear transformations, the projective transform is most general since all other transforms are sub-classes.

A specific property of the projective transform is that lines that are parallel in the source domain are now joining a common intersection point in the destination domain. This intersection point is called the *vanishing point*.

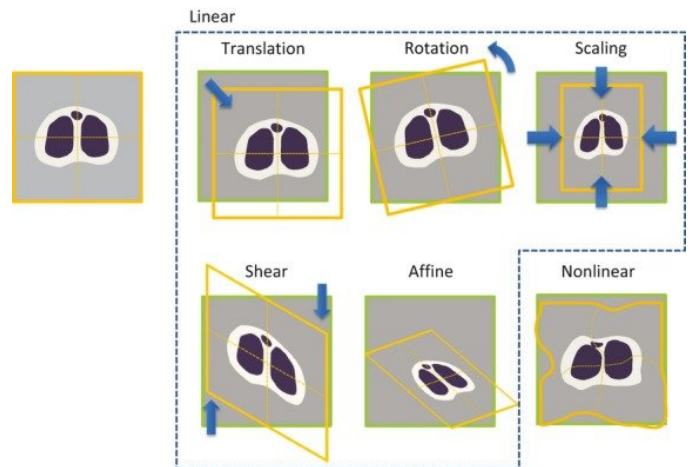


Figure 8: Overview of geometric transformation functions (Uchida et al)

2.2 Nonlinear registration

The transformation between the two point sets is not always assumed rigid and, in many cases, a more complex transformation has to be determined. In non-rigid registration correspondence between images cannot be achieved without taking localized deformations into account. For the localized deformations B-spline provide a good deformation model.

B-spline

A spline can be seen as a piecewise polynomial from which the pieces fit together at the joins.

For example (Ruekert et al), to find the optimal registration transformation $T: (x, y, z) \rightarrow (x', y', z')$ of for example a breast tissue you will need more than rigid or affine transformations

for motion correction as the breast is non-rigid.

A dynamic image sequence $I(x,y,z,t)$ at time which corresponds with $I(x',y',z',t_0)$ at timepoint t_0 . Therefore, a combined transformation T can be introduced to combine a global with a local transformation. The global transformation can be represented as an affine transformation including 12 DOFs to describe rotation, translation but also scaling and shearing:

$$T_{global}(x, y, z) = \begin{pmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \\ \theta_{31} & \theta_{32} & \theta_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} \theta_{14} \\ \theta_{24} \\ \theta_{34} \end{pmatrix}$$

Differences in age and patients determine local imaged deformations. For example, Free form deformation (FFD) based on B-spline can represent a good local deformation model, as an underlying mesh of control points can be manipulated to deform an object.

Points distributed on the surface of FFD deformable breast can be defined with the following image volume:

Image volume: $\Omega = \{(x, y, z) \mid 0 \leq x \leq X, 0 \leq y \leq Y, 0 \leq z \leq Z\}$

Ψ : a $n_x \times n_y \times n_z$ mesh of control points, $\Psi_{i,j,k}$ with uniform spacing δ

FFD: 3D tensor product of 1D cubic B-splines defined by:

$$T_{local}(x, y, z) = \sum_{l=0}^3 \sum_{m=0}^3 \sum_{n=0}^3 B_l(u) B_m(v) B_n(w) \Psi_{i+l, j+m, k+n}$$

With: $i = \left\lfloor \frac{x}{n_x} \right\rfloor - 1, j = \left\lfloor \frac{y}{n_y} \right\rfloor - 1$

$$u = \frac{x}{n_x} - \left\lfloor \frac{x}{n_x} \right\rfloor, v = \frac{y}{n_y} - \left\lfloor \frac{y}{n_y} \right\rfloor, w = \frac{z}{n_z} - \left\lfloor \frac{z}{n_z} \right\rfloor$$

Lth basis function B-spline model (B_l) defined by:

$$B_0(u) = \frac{(1-u)^3}{6}, B_1(u) = \frac{(3u^3 - 6u^2 + 4)}{6}$$

$$B_2(u) = \frac{(-3u^3 + 3u^2 + 3u + 1)}{6}, B_3(u) = \frac{u^3}{6}$$

The B-spline can be controlled locally by its control points. Changing a control point $\Psi_{i,j,k}$ will only affect the neighborhood of the control point. Model highly nonrigid deformations will be possible with small spacing of the control points.

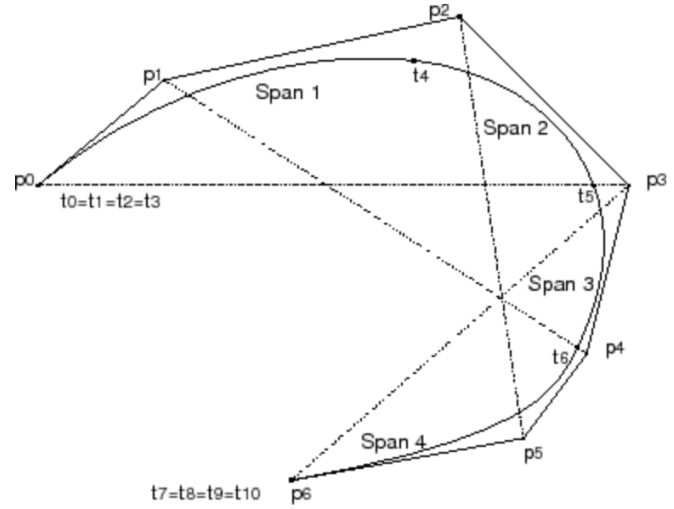


Figure 9: Example: 4th order cubic B-spline (example)

3 Matching criteria

3.1 Intensity based registration

Intensity or grey level-based registration methods match intensity patterns in each image. In general, they are using the full image content. The transformation between images is calculated using the pixel or voxel values. Mathematical and statistical criteria are used to execute this procedure.

3.1.1 Monomodal approach

The most basic similarity measure includes the sum of squared differences. This approach assumes that the images for registration are identical with exception of Gaussian noise. Another method correlation coefficient assumes a linear relationship between corresponding intensities of the images.

Similarity metrics

There are many metric methods used in image registration. Depending on the specific situation which metric suits best. The most used metrics are mentioned below:

- Mean squares difference metric
- Normalized cross correlation
- Mutual information

Mean squares difference metric describes two objects with intensity values in the same range mathematically as follows:

$$MSD = \frac{\sum_{i=0}^N (A_i - B_i)^2}{N}$$

A and B correspond the points in the registration process, and N is the number of points.

Normalized cross correlation is used to perform the registration of objects whose intensity values are related with a linear transformation. It represents the correlation between intensity values divided by the square rooted autocorrelation of the objects.

$$C(A, B) = \frac{\sum_{i,j} [a_{i,j} - \text{mean}(A)] x [b_{i,j} - \text{mean}(B)]}{\sqrt{\sum_{i,j} [a_{i,j} - \text{mean}(A)]^2} \sqrt{\sum_{i,j} [b_{i,j} - \text{mean}(B)]^2}}$$

3.1.2 Multimodal approach

In 1992 Woods introduced a registration measure for multimodal images, based on the assumption of correspondence between similar tissue type regions in two images.

The method is based on the difference in intensity characteristics from images acquired at different modalities.

The Shannon entropy for a joint distribution is described as in which H reflects the average information and a measure of the mean uncertainty reduction as function of probabilities:

$$H = - \sum p_{i,j} \log p_{i,j}$$

$p_{i,j}$ reflects the probability of co-occurrence of i and j.

Mutual information method is based on the dependence between two images, and how much information can be known during registration. Larger values are found in case of a better registration. Mutual information is obtained by subtracting the entropy of both sets from the joint entropy. Mathematically this is represented by the following equation:

$I(A,B)=H(B)-H(B|A)=H(A)+H(B)-H(A,B)$ in which:

$$H(A) = - \sum_a p_A(a) \log p_A(a)$$

$$H(B) = - \sum_b p_B(b) \log p_B(b)$$

$$H(A,B) = - \sum_{a,b} p_{A,B}(a,b) \log p_{A,B}(a,b)$$

Which results in:

$$I(A,B) = \sum_{a \in A} \sum_{b \in B} p_{AB}(a,b) \log \frac{p_{AB}(a,b)}{p_A(a) \cdot p_B(b)}$$

Mutual information is a dimensionless quantity and indicates how much a random variable is dependent on the other. A zero value for mutual information indicates independence between the variables.

3.2 Feature / key point based registration

3.2.1 Key points

Key points (synonyms: interest points, point features) are points, i.e. pixel positions, in the image plane that have desirable properties:

- They are well detectable, i.e. a point that is marked in one image can also easily be found in another image of the same scene.
- They are well locatable, i.e. the found position of the key point is well reproducible, and not sensitive to noise.
- They are well discernable, i.e. if a pixel marked as a key point in one image; it can easily be associated with the corresponding key point in another image.

In the vicinity of edges and lines you won't find points that meet these criteria as the position of a point along an edge or line is difficult to locate. Corners, spots, and blobs (larger spots) are better candidates for key points.

4 Optimization

Optimization refers to the procedure to improve the image similarity by adjusting the transformation. Choosing a good optimizer is important, as it reliably and quickly finds the best possible fit. A common used optimizer for registration is the Iterative closest point algorithm of Besl and McKay. This algorithm identifies the closest model point for each data point and finds the least square rigid body transformation relating the points.

As mentioned in figure 2 the optimization process refines the parameter values for optimal alignment of the moving image in an iterative process.

There are several algorithms which can be used for optimized and examples include:

- Gradient descent
- Quasi-Newton
- Nonlinear conjugate gradient
- Stochastic gradient descent methods
- Kiefer-Wolfowitz
- Robbins-Monro
- Adaptive stochastic gradient descent
- Preconditioned stochastic gradient descent
- Particle swarm optimization

The choice of an optimizer is dependent on the application and factors like: transformation, potential time constraints, and required accuracy of the registration.

A parametric optimization problem can describe the process of intensity-based registration between the fixed (I_F) and moving image (I_M):

$$\hat{\mu} = \operatorname{argmin}_{\mu} \vartheta(I_F, I_M \circ T_{\mu})$$

In which:

$T_{\mu}(\mathbf{x})$: coordinate transformation parametrized by μ .

ϑ : dissimilarity measure

The following formula is used to solve the registration problem

$$\mu_{K+1} = \mu_K - \gamma_k \mathbf{d}_k$$

K : iteration number

γ_k : Step size at iteration k

$\mathbf{d}_k$: search direction

Gradient descent directions are commonly used for the direction of search. Stochastic variants are preferred for their efficient implementation and less computation costs.

5 Sampling

In an iterative process (Figure 1) the loop takes into account all voxels of the fixed image. In practice, also a subset can be used and selected based on the grid or randomly. This is an advantage due to the high computational power that is needed when large datasets are considered.

The choice to resample is dependent on the accuracy which is demanded and the computational complexity that can be justified.

6 Practical information

There are several software workpackages which could be used for registration:

- Matlab – Mathworks, for example to pick the correct optimizer and metric configuration (<https://www.mathworks.com/help/images/registering-multimodal-mri-images.html>)
- Elastix

- Plastimatch
- Worldmatch
- Mevislab
- 3D slicer
- ITKsnap

7 Example: Estimation of the registration transform parameters

Suppose that corresponding key points have been found in both images. Also, suppose that a transformation type, say, $H\{\}$ has been adopted, that maps key points from the second image (the moving image) to the first image (the reference image):

$$\mathbf{x}_1 = H\{\mathbf{x}_2\} \quad (1)$$

In fact, (1) is the same as **Error! Reference source not found.**, but now written in vector form. A special case of $H\{\}$ is the homography $\mathbf{x}_1 = \mathbf{H}\mathbf{x}_2$ as in **Error! Reference source not found.** In general, the transformation will depend on a number of parameters, which we collect in a parameter vector α . In case of the homography, the matrix $\mathbf{H}$ contains 8 independent parameters, so the vector α will be 8-dimensional here. But if other types of transformations are adopted, the number of parameters could be much more, or much less.

The problem addressed in this section is how to estimate the parameter vector α from the available matched key points?

To elucidate the dependence of $H\{\}$ on α , we will write $H_{\alpha}\{\}$. The estimation of α can then be formulated as an optimization problem:

$$\hat{\alpha} = \operatorname{argmin}_{\alpha} d(X_1, H_{\alpha}\{X_2\}) \quad (2)$$

X_1 and X_2 are the sets of key points in the first and second image. $H_{\alpha}\{X_2\}$ is the set X_2 transformed by $H_{\alpha}\{\}$, and $d(\cdot, \cdot)$ is a distance measure between the two sets. A well-known measure is the SSD, sum of squared differences:

$$d(X_1, H_{\alpha}\{X_2\}) = \sum_{n=1}^N \|\mathbf{x}_1(n) - H_{\alpha}\{\mathbf{x}_2(\hat{m}(n))\}\|^2 \quad (3)$$

The optimization strategy works out differently for the different transformation types. If $H_{\alpha}\{\}$ is a similarity transform, then there are 4 parameters: the change of scale s , the rotation angle φ , and the translation vector $\mathbf{t}$. These parameters can be estimated by the so-called Procrustes

algorithm, which will be dealt with in a follow-up course. Alternatively, the parameters can also be estimated by the set of scales $\sigma_1(n)$ and $\sigma_2(m)$, and the orientations $\theta_1(n)$ and $\theta_2(m)$, along with the positions of the key points.

In case the transformation type is a homography, the parameters can be estimated by the so-called *eight point algorithm*; see below. If the transformation type is a non-linear type, the optimization must usually be done by means of numerical optimization.

7.1 The inhomogeneous solution

If the transformation type is a homography, the solution can be obtained by solving the inhomogeneous equations which turns out to be linear in α . The homogenous equation is given in **Error! Reference source not found.**. Expanding this, using **Error! Reference source not found.**, yields:

$$\begin{aligned} \alpha x_1 &= Ax_2 + By_2 + C \\ \alpha y_1 &= Dx_2 + Ey_2 + F \\ \alpha &= Gx_2 + Hy_2 + 1 \end{aligned} \quad \text{or:} \quad (4)$$

$$\begin{aligned} (Gx_2 + Hy_2 + 1)x_1 &= Ax_2 + By_2 + C \\ (Gx_2 + Hy_2 + 1)y_1 &= Dx_2 + Ey_2 + F \end{aligned}$$

Collecting the terms with the unknown parameters gives the inhomogeneous, linear equations for the parameters:

$$\begin{aligned} x_2A + y_2B + C - x_1x_2G - x_1y_2H &= x_1 \\ x_2D + y_2E + F - y_1x_2G - y_1y_2H &= y_1 \end{aligned} \quad (5)$$

Transforming this in vector-matrix notation:

$$\begin{bmatrix} x_2 & y_2 & 1 & 0 & 0 & 0 & -x_1x_2 & -x_1y_2 \\ 0 & 0 & 0 & x_2 & y_2 & 1 & -y_1x_2 & -y_1y_2 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \\ G \\ H \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad (6)$$

which is of the form:

$$\mathbf{A}\alpha = \mathbf{b} \quad (7)$$

These are effectively 2 equations for 8 unknown parameters, and it is obtained with just one pair of matching key points. If 3 other key points are added, the matrix $\mathbf{A}$ becomes 8×8 -matrix and $\mathbf{b}$ an 8D vector. Under

the condition that the 4 key points do not show any collinearity, the matrix $\mathbf{A}$ is invertible, so that the solution becomes:

$$\hat{\alpha} = \mathbf{A}^{-1}\mathbf{b} \quad (8)$$

If more than 4 key points are available, which is usually the case, (7) is overdetermined, and a least squares solution of the problem is provided by the pseudo-inverse:

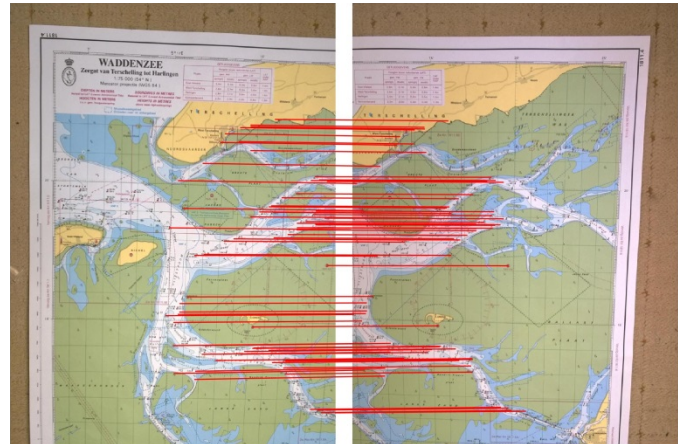
$$\hat{\alpha} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b} \quad (9)$$

7.2 Robust estimation

If the homography is estimated with a set of matching pairs but also with a few wrong matches, then the resulting homography would be very bad, even if the wrong matches forms a very small minority. To be robust against these so-called outliers, robust estimation is needed.

One way to achieve this is to adopt a so-called *robust error norm* instead of the SSD as in (3). Another possibility is to use an 'error and trial' approach. The latter is implemented in the RANSAC algorithm. This algorithm tries to find a solution by first selecting a small subset of the matching pairs, hoping that this subset does not contain outliers. The parameters are estimated with this subset. Then, the whole set of key points of the second image is transformed with this found solution. Each transformed key point is compared with the matching key point in the first image to see whether it fits. Those key points which really fits (with some tolerance) are taken together in a so-called consensus set. If the consensus is too small, the trial is abandoned, and another trial is started. If not, then the final estimate is obtained by using the consensus set.

As an example, RANSAC is applied to the previous example yielding a consensus set (inliers) of 71 matching pairs:



The corresponding homography is given by:

$$\mathbf{H} = \begin{bmatrix} 0.939 & 0.062 & -287.6 \\ -0.045 & 0.962 & 24.8 \\ -0.000 & 0.000 & 1 \end{bmatrix}$$

which shows that the homography is virtually an affine transform.

8 Geometrical transforms in Matlab

Geometrical transforms are implemented in two different toolboxes. The image processing toolbox supports the function `imtransform` with a number of supporting functions. However, this set of functions has become obsolete with introduction of the computer vision toolbox that contains the function `imwarp` with supporting functions.

Below is the resulting of applying the suit of functions from the computer vision toolbox. The Matlab code with which this image is produced is:

```
%% find SURF key points
points1 = detectSURFFeatures(im1);
points2 = detectSURFFeatures(im2);

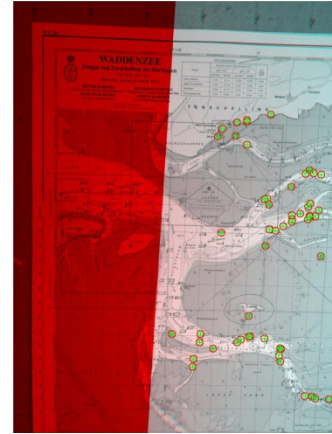
%% find SURF features
[features1,validPoints1]= ...
    extractFeatures(im1,points1);
[features2,validPoints2]= ...
    extractFeatures(im2,points2);

%% match them
indexPairs12 = ...
    matchFeatures(features1,features2);
p12 = validPoints1(indexPairs12(:,1),:);
p21 = validPoints2(indexPairs12(:,2),:);

%% estimate the transform
[tform, iP2, iP1, status] = ...
    estimateGeometricTransform(p21,p12, 'projective');

%% Q9 tranform the image
[im2t, im2ref] = imwarp(im2g, tform, 'OutputView',...
    imref2d(size(im1g)), 'FillValues',0);
iP2t = transformPointsForward(tform, iP2.Location);
showMatchedFeatures(im1g, im2t, iP1, iP2t);
```

In the function `imwarp`, the second image is transformed with the first image as a spatial reference for the output of the transformed image. This makes sure that the pixel positions of the output image are comparable with the pixel positions of the first image.



The function `showMatchedFeatures` is especially designed for comparing registered grey level images. If pixels are grey, the images are perfectly aligned. If there is a difference, then this shows up in red or in blue. The matching key points are shown in red and green.

9 References / background

- Goldman, Pyramid Algorithms, 2003
- Yousefi, Two-stage registration of substructures in magnetic resonance brain images, 2009
- Patrikalakis-Maekawa, B-spline Approximation and the Boor Algorithm, MIT, <https://web.mit.edu/>
- Uchida et al. Image processing and recognition for biological images, 2013.
- Goshtasby, Image registration – principles, tools and methods, 2012.